Chemical Engineering | Semester II

2071 — Fundamentals of Engineering Chemistry

Credits: 4.5 | Contact Hours: 90 | Curriculum: REV2026

Comprehensive study of Electrochemistry, Surface Chemistry, and Organic Chemistry fundamentals for Chemical Engineers.

Download Study Notes

Official Source Declaration

This lesson is constructed based on the official **SITTTR Kerala Revision 2026 Syllabus** for Course Code **2071 (Fundamentals of Engineering Chemistry)**. The content covers all modules, subtopics, and practical requirements as specified in the official document.

Note on Contact Hours: The syllabus specifies 90 total contact hours. The detailed subtopic breakdown provides approximately 45 Lecture hours and 39 Practical hours (Total 84). This lesson adheres to the detailed topic distribution provided in the official syllabus.

Course Dashboard

Attribute	Details
Program	Chemical Engineering
Course Type	Practicum (Theory + Practical)

Teaching Scheme	3:0:3:0 (L:T:P:R)
CIE Marks	40 (20 Theory + 20 Practical)
ESE Marks	100 (60 Theory + 40 Practical)
Pre-requisite	2003B - Chemistry for Technical Practices

How to Study This Lesson

- **Theoretical Foundation:** Start with the lecture topics in each module to understand the principles.
- **Practical Integration:** This is a Practicum course. Link the experiments (e.g., Daniel Cell construction) directly to the theory (Electrochemistry).
- **Visual Learning:** Use the SVG diagrams to memorize cell structures and phase diagrams.
- **Malayalam Support:** Refer to the `ml-note` blocks if you find technical English concepts difficult to grasp initially.

Course Outcomes (CO)

CO	Description	Bloom's Level
CO1	Apply electrochemistry and corrosion knowledge to industrial challenges.	Apply
CO2	Apply adsorption mechanisms, colloids, and heterogeneous systems.	Apply
CO3	Apply fundamental organic chemistry to solve elementary problems.	Apply
CO4	Explore industrially important organic compounds and their reactions.	Apply

Syllabus Coverage

Module	Title	L Hours	P Hours
1	Electrochemistry, Corrosion & Battery Technologies	11	9

Module	Title	L Hours	P Hours
2	Surface Chemistry, Colloids & Heterogeneous Systems	11	9
3	Chemical Bonding, Carbon & Purification Methods	11	9
4	Important Organic Compounds & Reactions	12	12

Learning Roadmap

Step 1: Electron Transfer (Electrochemistry) & Material Degradation (Corrosion).

Step 2: Interface Phenomena (Adsorption) & Complex Mixtures (Colloids).

Step 3: Molecular Architecture (Bonding, Hybridization) & Separation Science.

Step 4: Synthetic Pathways for Industrial Chemicals (Hydrocarbons, Alcohols, Phenols).

Module I

Electrochemistry, Corrosion and Emerging Battery Technologies

Lecture: 11 Hours Practical: 9 Hours

1.1 Electrochemistry Fundamentals

Electrochemistry deals with the relationship between electrical energy and chemical change.

- **Oxidation:** Loss of electrons.

- **Reduction:** Gain of electrons.
- **Redox Reaction:** Simultaneous oxidation and reduction.

> **Malayalam support:** Oxidation □□□□□□ □□□□□□□□□□□ □□□□□□□□□□□□□□□□ □□□□□□□□□□□□. Reduction □□□□□□ □□□□□□□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□. □□ □□□□□□ □□□ □□□□ □□□□□□□□□□□ □□□□□ Redox reaction □□□□□ □□□□□□□□□□□□.

1.2 Conductors and Electrolytes

Type	Mechanism	Example
Metallic Conductors	Flow of electrons	Copper, Aluminium
Electrolytic Conductors	Movement of ions	NaCl solution, H2SO4

Strong Electrolytes: Completely ionize (HCl, NaOH).
Weak Electrolytes: Partially ionize (CH3COOH, NH4OH).

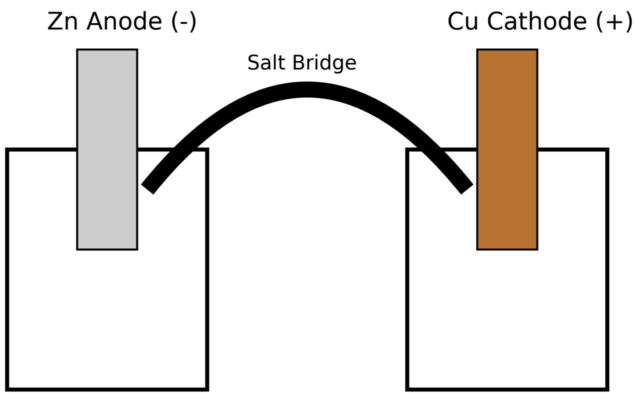
1.3 Faraday's Laws of Electrolysis

First Law: $m = ZIt$

Second Law: $m1/m2 = E1/E2$

Given: Current (I) = 2A, Time (t) = 10 mins (600s), Z for Copper = 0.000329 g/C.
Required: Mass of copper deposited (m).
Calculation: $m = Z \times I \times t = 0.000329 \times 2 \times 600 = 0.3948$ g.
Answer: 0.3948 grams.

1.4 Electrochemical Cells (Daniel Cell)



Simplified Daniel Cell Diagram

1.5 Corrosion and Prevention

Corrosion is the gradual destruction of metals by chemical/electrochemical reaction with the environment.

- **Barrier Protection:** Anodic/Cathodic coatings, Anodizing.
- **Cathodic Protection:** Sacrificial Anode method (using Zn or Mg to protect Fe).

1.6 Emerging Battery Technologies

- **Li-ion:** High energy density, used in phones/EVs.
- **Na-ion:** Cheaper alternative to Li-ion using abundant Sodium.
- **Solid State Battery:** Uses solid electrolyte instead of liquid; safer and higher capacity.

Module II

Surface Chemistry, Colloids and Heterogeneous Systems

Lecture: 11 Hours   Practical: 9 Hours

2.1 Adsorption

Adsorption is a surface phenomenon where molecules accumulate on the surface of a solid.

Feature	Physisorption	Chemisorption
Force	Weak Van der Waals	Strong Chemical Bonds
Specificity	Non-specific	Highly specific
Reversibility	Reversible	Irreversible

2.2 Colloids

Particles sized between 1nm and 1000nm.

- **Tyndall Effect:** Scattering of light by colloidal particles.
- **Brownian Movement:** Random zig-zag motion of particles.
- **Electrophoresis:** Movement of particles under an electric field.

> **Malayalam support:** Colloids □□□□□□□ □□□□□ □□□□□□ □□□□□□□ (1-1000nm) □□□□□□□□ □□□□□□□□□□□□□. □□□□□□□□ □□□□□□□□ □□□□□□□□□□□□□□□□□ □□□□ □□□□□□□□□□□□□□□□□□□□ Tyndall Effect □□□□□□ □□□□□□□□□□□□□□.

2.3 Heterogeneous Systems (Phase Rule)

Gibb's Phase Rule: `F = C - P + 2`

For a Water System (One component, C=1):

- Triple Point: P=3, F = 1 - 3 + 2 = 0 (Invariant).
- Curves: P=2, F = 1 - 2 + 2 = 1 (Univariant).
- Areas: P=1, F = 1 - 1 + 2 = 2 (Bivariant).

Module III

Chemical Bonding, Carbon and Purification Methods

Lecture: 11 Hours   Practical: 9 Hours

3.1 Chemical Bonding

- **Ionic:** Complete transfer of electrons (e.g., NaCl).
- **Covalent:** Sharing of electrons (e.g., HF).
- **Coordinate:** One atom provides both electrons for sharing (e.g., $NH4+$).

3.2 Unique Nature of Carbon

- **Tetravalency:** Carbon has 4 valence electrons.
- **Catenation:** Ability to form long chains or rings with other carbon atoms.
- **Allotropes:** Diamond, Graphite, Fullerenes.

3.3 Hybridization

Type	Geometry	Example
sp^3	Tetrahedral	Methane (CH4)
sp^2	Trigonal Planar	Ethene (C2H4)
sp	Linear	Ethyne (C2H2)

3.4 Purification of Organic Compounds

- **Sublimation:** Solid to gas directly (e.g., Camphor).
- **Distillation:** Based on boiling point differences.
- **Chromatography:** Based on differential adsorption (TLC, Column).

Module IV

Preparation and Chemical Reactions of Organic Compounds

Lecture: 12 Hours   Practical: 12 Hours

4.1 Hydrocarbons

- **Methane:** Decarboxylation of Sodium Acetate with Sodalime.
- **Ethane:** Wurtz reaction (Chloromethane + Na in dry ether).
- **Ethyne:** Calcium carbide + Water.

4.2 Important Reactions

Markovnikov’s Rule: In the addition of HBr to an unsymmetrical alkene, the H attaches to the carbon with more hydrogen atoms.

Friedel-Crafts Reaction: Alkylation or Acylation of Benzene using AlCl3 catalyst.

> **Malayalam support:** Markovnikov’s Rule □□□□□□□, □□□ unsymmetrical alkene-□□□□□□ HBr □□□□□□□□□□□□□, □□□□□□ □□□□□□□ □□□□ □□□□□□□□□□□□□□ H □□□□ □□□□□□□□□.

4.3 Alcohols, Phenols and Acids

- **Ethanol:** Prepared by fermentation of molasses. Used as fuel and solvent.
- **Phenol:** Prepared by Dow's process. Undergoes Nitration and Kolbe reaction (to form Salicylic acid).
- **Ethanoic Acid:** Prepared from CO2 using Grignard reagent. Undergoes HVZ reaction.

Suggested Student Activities

Module I & II

- Report on Hydrogen Economy.
- Assignment on Battery recycling.
- Video on Adsorption in water purification.
- Demonstrate Tyndall effect at home.

Module III & IV

- Assignment on purification methods.
- Model building for Hybridization.
- Identify household organic chemicals.
- Analyze constituents of sanitizers/perfumes.

Formula Bank

Faraday's 1st Law

$$m = ZIt$$

Gibb's Phase Rule

$$F = C - P + 2$$

Freundlich Isotherm

$$x/m = kP^{1/n}$$

Visual Library

Daniel Cell

Essential for understanding electrochemical potential and salt bridge function.

Water Phase Diagram

Shows the relationship between Ice, Water, and Vapor at different T and P.

Assessment Methodology

Component	Weightage	Details
CIE (Theory)	20 Marks	Series tests, Attendance, Assignments.
CIE (Practical)	20 Marks	Lab performance, Record, Viva.
ESE (Theory)	60 Marks	2.5 Hours exam covering all modules.
ESE (Practical)	40 Marks	3 Hours internal lab exam.

Module-wise Questions

Module 1: Distinguish between Metallic and Electrolytic conductors.

Metallic conductors conduct electricity via flow of electrons without chemical change (e.g., Cu). Electrolytic conductors conduct via movement of ions accompanied by chemical decomposition (e.g., NaCl soln).

Module 2: What is the Tyndall Effect?

It is the scattering of a beam of light by colloidal particles, making the path of light visible. This happens because colloidal particles are large enough to scatter light.

Module 3: Explain Catenation in Carbon.

Catenation is the unique property of carbon atoms to form stable covalent bonds with other carbon atoms, resulting in long chains (linear or branched) and rings.

Module 4: State Markovnikov's Rule with an example.

Rule: In addition of HX to an unsymmetrical alkene, the H attaches to the carbon with more H atoms. Example: Propene + HBr → 2-Bromopropane (Major product).

Practice Model Paper

Note: This is a practice paper, not an official examination paper.

Part A (1 Mark each)

1. Define Oxidation.
2. What is a Triple Point?
3. Name an allotrope of Carbon.
4. Write the formula of Methane.

Part B (3 Marks each)

5. Explain the sacrificial anode method of corrosion prevention.
6. Differentiate Physisorption and Chemisorption.
7. Explain sp3 hybridization in Methane.

Part C (7 Marks each)

8. Describe the construction and working of a Daniel Cell with a neat diagram.
9. Explain the Phase diagram of the Water system using Gibb's Phase Rule.

Quick Revision

Key Points:

- Anode = Oxidation; Cathode = Reduction.
- F = C - P + 2.
- Colloids: 1nm to 1000nm.
- Carbon: 4 bonds, long chains.

Exam Checklist:

- Faraday's Law problems.
- Daniel cell diagram.
- Water phase diagram.
- IUPAC Naming rules.

Glossary

Electrolyte
: A substance that produces an electrically conducting solution when dissolved in a polar solvent.

Adsorbate
: The substance that gets adsorbed on the surface.

Hybridization
: The concept of mixing atomic orbitals to form new hybrid orbitals.

Isotherm
: A curve showing the relationship between pressure and amount adsorbed at constant temperature.

References

- Principles of Physical Chemistry - Puri, Sharma & Pathania (Vishal Publishing).
- Organic Chemistry - Bahl & Arun Bahl (S. Chand).
- Textbook of Physical Chemistry - P.L. Soni.
- NPTEL Course: 104106119.